

Project Description – Project Proposals

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Keep the breath in mind: Respiratory interoception in the developing brain

Project Description

1 Starting Point

Life starts with a breath and ends with one. We can live without food and water for days but we can only survive minutes without breathing. Despite being a vital motor task that is performed on average 20000 times a day, it is surprising that little is known about how breathing interacts with neural, sensorimotor and bodily processes to shape the sense of self.

Interoception, which is traditionally defined as the perception of the physiological states of one's body, highlights the embodied nature of the self by situating the agent in a body that is situated in the outside world at the present moment. Earlier, we have argued that one key aspect of minimal self, which can be described as the phenomenal experience of being here and now, originates from the perception and interpretation of interoceptive signals (Fotopoulou & Tsakiris, 2017; Seth et al., 2012). The embodied nature of interoception alludes to the argument that the self is not a static, abstract concept, but it is an embodied, dynamic phenomenon constructed on the basis of bodily, sensorimotor and interoceptive experiences.

Breathing lies at the intersection between interoception, exteroception and proprioception. Unlike any other interoceptive modality, it can easily be brought into awareness, and more importantly, it can directly be controlled. The uniqueness of breath as an actively malleable interoceptive modality makes it a perfect candidate to examine the mechanisms underlying the minimal self. In this proposal, we set the stage to investigate whether and how respiratory interoception interacts with neural and bodily processes and how active modulation of the breath elicits short-term and long-term changes in the developing brain.

1.1 State of the art and preliminary work

The interplay between respiration and physiological functions such as cardiac activity has long been investigated both in adults (Sroufe, 1971) and in children (Sroufe & Morris, 1973). For example, slowing down the breathing tempo to 6 breaths per minute (i.e. 0.1 Hz), which is known as slow-paced breathing, increases respiratory sinus arrhythmia (RSA), a phenomenon in which intervals between two consecutive heartbeats shorten during inspiration and prolong during expiration. It has been widely established that breathing slowly increases heart rate variability (HRV), defined as the variation in the time intervals between adjacent heartbeats, which is considered to be an indicator of a healthy regulatory capacity (Shaffer et al., 2014).

One fundamental yet underexplored aspect of brain and body interactions is the link between the respiratory and the nervous systems. A growing body of evidence from animal and clinical research showed that neural oscillations entrain to respiration, and respiration modulates cognitive and affective functions (Heck et al., 2017; Herrero et al., 2018; Tort et al., 2018; Zelano et al., 2016). For example, high frequency oscillations such as gamma (40-150 Hz), which are known to govern attention, memory and sensory perception, are coupled to respiration (Heck et al., 2017; Tort et al., 2018). Emerging evidence from healthy adults showed that respiration modulates the oscillatory activity during a motor task as well as at rest (Kluger & Gross, 2020a; 2020b).

Research further revealed how attention to the breath and the active modulation of it influence neural oscillations in the brain. In an intracranial electroencephalography (iEEG) study with epileptic patients, Herrero and colleagues (2018) demonstrated that gamma band activity coupled to the respiration, both when the participants attended to their breaths and when they actively modulated them. Using magnetoencephalography (MEG), Hsu and colleagues (2020) showed that slow as compared to normal breathing elicited phase adjustments in neural oscillatory activity across widespread brain areas. Evidence, thus, suggests that respiration, as an interoceptive modality, interacts with sensorimotor and bodily processes to organize cortical activity.

The interplay of slow and fast cortical rhythms might reflect the relay of bottom-up sensory information and top-down predictions. In other words, a hierarchy of neural rhythms may be orchestrating brain-body interactions through an active inference mechanism, which mediates information between primary afferent pathways and higher cortical regions (Corcoran et al., 2018). More specifically, respiratory information from pulmonary and nasal pathways is propagated up to cortical areas responsible for the integration of interoceptive and exteroceptive signals, and vice versa, top-down predictions issued by cortical regions then regulate physiological responses. The connection between primary sensory regions and the cortex further engages areas associated with emotion, arousal, attention and memory. Consequently, regulating the breath means regulating this neural circuitry, and the psychophysiological processes associated with it (Boyadzhieva & Kayhan, 2020).

The overarching purpose of a respiration-based regulatory mechanism is the adaptation to a precarious environment. We have argued that through active inference, that is, changing the breathing rhythm, interoceptive prediction errors (PE) can be minimized. As respiration-related sensory evidence is recontextualized, the interoceptive predictions are altered. For example, in the case of slowing down the breathing rhythm when anxious, they become aligned with a relaxed interoceptive state. Whereas active modulation of the breath may induce rapid positive changes in the internal states, it is an intriguing question whether repeated experience with active modulation of the breath (i.e. training) can also elicit long-lasting changes in interoceptive generative models (Boyadzhieva & Kayhan, 2020).

Evidence suggests that breath training modulates physiological and neural processes in healthy adults (Laborde et al., 2019; for a review, Zaccaro et al., 2018). Studies in adult mindfulness literature show that with repeated experience, one can readily attend to bodily signals such as the breath to trigger regulatory strategies such as slow-paced breathing in response to environmental stressors (Farb et al., 2013; Weng et al., 2020). These findings are in line with the widely-accepted role of interoceptive processing on emotion regulation and mental health (Barrett et al., 2016; Critchley & Garfinkel, 2017; Murphy et al., 2017). Taken together, we argue that controlled breathing can be used as an easily accessible self-regulation tool to modulate neural and bodily processes fostering cognitive and emotional well-being.

Despite the growing body of evidence on how breathing interacts with neural and bodily processes in animals and adults, it is entirely unknown how these interactions work in the developing human brain. First, up to this date, there has been no research on how respiration

modulates neural oscillations in development. Second, changes in the neuronal activity both when attending to the respiratory signals and when actively modulating them is yet to be discovered. Third, the question of whether and how repeated experiences with active modulation of the breath induce long-term changes in neural, bodily and interoceptive processes is unexplored. Therefore, we believe that the proposed project will greatly contribute to the literature by pioneering this research line. Considering the scarcity of research on the development of interoception (Murphy et al., 2017), we are certain that the work we started in the first phase, which we aim to further develop in the second period, will be of crucial value to the Priority Program filling an important knowledge gap in the literature.

In the first phase of the project, which started in May 2019, we investigate the emerging sense of self in infancy, operationalized through interoceptive sensitivity, in relation to early interactions with the primary caregiver. In this project, we measure interoceptive sensitivity in infants and their mothers. With a multilevel hyperscanning approach, we examine the interaction between the mother and the infant to define behavioral, physiological and neural mechanisms that contribute to the interoceptive sensitivity of the infants. We initially planned to monitor the development of interoceptive sensitivity, when the infant is at 3, 9, and 18 months of age. Due to the challenges arising from the Covid-19 pandemic (e.g. closure of universities, lack of 3-month-old infants in the databases due to the suspension of recruitment in hospitals, infeasibility of EEG measurements due to prolonged durations of close contact during the application of the caps etc.), we had to adjust our initial plans. Although we have successfully implemented all of the experimental paradigms in the original proposal, we continued testing with the paradigms that were approved given the testing regulations in the University of Vienna, where the data is collected. Except for the EEG measurements, we are testing infants and their mothers using several behavioral and physiological measures such as electrocardiogram (ECG) and respiratory measurements. Whereas we successfully use dual ECG to measure cardiac synchrony between the mothers and their infants, as planned, we examine the neural synchrony between the interaction partners using dual functional near-infrared spectroscopy (fNIRS), instead of dual EEG, due to the feasibility of application. As for the age group, we have started the data collection with 9-month-old infants and their mothers, which is intermittently ongoing up to this date. We plan to test the infants again when they are at 18 months of age. Below, we will present some preliminary findings from infant interoceptive sensitivity studies, which have been pre-registered in the AsPredicted platform.

In one study, we investigate the behavioral markers of cardiac interoceptive sensitivity in infants by using an innovative paradigm developed by our external collaborator in the project (Maister, Tang, & Tsakiris, 2017). The iBEAT measures infants' sensitivity to their own heartbeats. In this task, infant-friendly images (e.g. stars or clouds with faces), which move either in sync or out of sync with the infant's own heartbeat, are presented to the infant while their looking time to each stimulus is measured. We hypothesized that if infants distinguish their heartbeat as an indication of sensitivity to their interoceptive signals, they will look longer at the asynchronous trials as compared to the synchronous trials, as the asynchronous stimuli will violate their expectations on their own interoceptive signals. Preliminary trends in the data indicate that infants at 9 months of age ($N=6$) look longer at the images that are in sync ($M=7507.64$ ms, $SD=1030.88$ ms) with their cardiac signals as compared to asynchronous images ($M=5614.44$ ms, $SD=927.15$ ms). More data is being acquired to better shed light on the cardiac interoception in infancy.

In another experiment, we examine *respiratory* interoceptive sensitivity in early development. Accordingly, we successfully developed the iBREATH paradigm, in which we present infants with animated characters on the screen that expand and shrink in sync or out of sync with their respiratory rate, while we measure their looking times at the images. This is the first-ever paradigm designed to measure infant respiratory interoception. In line with the predictions on cardiac interoception, we expected infants to look longer at the images that are out of synchrony with their respiratory rate, if they recognize their respiratory signals. Preliminary data demonstrate that infants at 9 months of age ($N=10$) tend to look almost equally long at the synchronous ($M=6348.39$ ms, $SD=728.29$ ms) and asynchronous ($M=6385.97$ ms,

$SD=593.94$ ms) images. Data collection is ongoing to better understand respiratory interoception in infancy.

In the proposed interdisciplinary project, we aim to further shed light on, neural, behavioral and computational mechanisms underlying interoceptive models in development. Combining cutting-edge empirical methods with robotics simulations, we will run a thorough investigation into how respiratory interoception interacts with neural and bodily processes and how active modulation of respiratory interoceptive signals alters interoceptive predictions and emotional experience. Our interdisciplinary aims resonate with the ultimate goals of the Priority Program integrating a mechanistic explanation on neural, behavioral and computational dynamics of respiratory interoception in development into a coherent model of minimal self. With this focus, our proposal mainly addresses the following key questions of the Priority Program:

- To which extent is the self plastic?
- To which degree does the sensorimotor impact on the self rely on or interact with internal, endogenous constraints?
- What are the mechanisms and prerequisites that allow an agent to develop a self?

In the first work package, we will investigate children's sensitivity to their instantaneous respiratory signals. In two experiments, we will measure the interplay between neural and respiratory dynamics at rest as well as in interoceptive attention tasks. This work package will lay the foundation for the other two work packages. In the second work package, we will focus on the behavioral, neural and computational mechanisms of active modulation of the breath. With a biofeedback game, we will test whether and how children learn to perform slow-paced breathing. Using computational modeling, we will formalize the dynamic mechanisms underlying active modulation of the breath in terms of short-term and long-term prediction error (PE) minimization. In the third work package, we will further investigate the role of experience with active modulation of respiratory signals on neural and interoceptive processes. Using a training procedure, we will test the prolonged effects of repeated sensorimotor learning (i.e. slow-paced breathing) on neurorespiratory dynamics as well as on interoceptive attention. We plan to test 6- to 8-year-old children for the second phase of the Priority Program. The first reason to focus on middle childhood is to further fill the knowledge gap in the literature with respect to the development of interoception (Murphy et al., 2017), particularly respiratory interoception. The second reason to choose this age group is that we aim to investigate the active modulation of interoceptive signals through biofeedback. This is the youngest age group that allows implementing such a procedure.

1.2 Project-related publications

1.2.1 Articles published by outlets with scientific quality assurance, book publications, and works accepted for publication but not yet published.

1. Nguyen, P. D. H., Georgie, Y. K., **Kayhan, E.**, Eppe, M., Hafner, V. V., & Wermter, S. (accepted). Sensorimotor representation learning for an "active self" in robots: A model survey. *Künstliche Intelligenz*.
2. Köster, M., **Kayhan E.**, Langeloh, M., & Hoehl, S. (2020). Making Sense of the World: Infant learning from a predictive processing perspective. *Perspectives on Psychological Science*, 15(3),562-571. doi:10.1177/1745691619895071. (open access)
3. **Kayhan, E.**, Meyer, M., O'Reilly, J. X., Hunnius, S., & Bekkering, H. (2019). Nine-month-old infants update their predictive models of a changing environment. *Developmental Cognitive Neuroscience*, 38, 100680. doi: 10.1016/j.dcn.2019.100680. (open access)
4. **Kayhan, E.**, Hunnius, S., O'Reilly, J. X., & Bekkering, H. (2019). Infants differentially update their internal models of a dynamic environment. *Cognition*, 186, 139-146. doi: 10.1016/j.cognition.2019.02.004. (open access)
5. **Kayhan, E.**, Heil, L., Kwisthout, J., van Rooij, I., Hunnius, S., & Bekkering, H. (2019). Young children integrate current observations, priors and agent information to predict others' actions. *PLOS One*, 14(5), e0200976. doi: 10.1371/journal.pone.0200976. (open access)

1.2.2 Other publications, both peer-reviewed and non-peer-reviewed

1. Nguyen, T., Schleihau, H., Kungl, M., **Kayhan, E.**, Hoehl, S., & Vrticka, P. (2021). Interpersonal neural synchrony during father-child problem solving: An fNIRS hyperscanning study. *Child Development*. doi: <https://doi.org/10.1111/cdev.13510>. (open access)
2. Michel, C., **Kayhan, E.**, Pauen, S., & Hoehl, S. (2021). Effects of reinforcement learning on gaze following of gaze and head direction in early infancy: An interactive eye-tracking study. *Child Development*. doi: <https://doi.org/10.1111/cdev.13497>. (open access)
3. Boyadzhieva, A., & **Kayhan, E.** (2020). Keeping the breath in mind: Respiration, neural oscillations and the free energy principle. *PsyArXiv*. <https://doi.org/10.31234/osf.io/8fm2>.
4. Nguyen, Q.T., Schleihau, H., **Kayhan, E.**, Matthes, D., Vrticka, P., & Hoehl, S. (2020) Neural synchrony in mother-child conversation: Exploring the role of communicative features. *Social Cognitive and Affective Neuroscience*. nsaa079. doi: 10.1093/scan/nsaa079. (open access)
5. Nguyen, Q.T., Schleihau, H., **Kayhan, E.**, Matthes, D., Vrticka, P., & Hoehl, S. (2020). The effects of interaction quality on neural synchrony during mother-child problem solving. *Cortex*, 124, 235-249. doi:10.1016/j.cortex.2019.11.020. (open access)

1.2.3 Patents

N/A

2 Objectives and work programme

2.1 Anticipated total duration of the project

The project will run for three years. In the first year, we plan to implement all of the experimental paradigms in the work packages. As we plan to use the same equipment that was purchased in the first phase of the project, we expect to start the implementation of the experimental paradigms right after the start of the project. We estimate 6 months for the implementation. Immediately after the implementation phase, we will start pilot measurements with children to further optimize the procedures for which we allocate 6 to 8 weeks. **Given that we plan to test early school-age children, the measurements will take place between 16:00-18:00 on weekdays. Under normal circumstances, depending on the availability of the testing facilities (i.e. number of participants in the databases, weekly allowance for lab space, etc.), we anticipate the entire data collection to take up to 9 months.** In parallel to the data collection, we will build data analysis pipelines. We plan to finalize data analyses and work on the collaborative studies at the end of the second year of the project. In the last 12 months of the project, we plan to finalize the collaborative projects and focus on writing the manuscripts and publishing.

Timeline of the proposed interdisciplinary project

May 2022	-	November 2022	Implementation of the experimental paradigms, setting up the experiments
December 2022	-	February 2023	Pilot measurements, data quality assurance
March 2023	-	December 2023	Data collection for all work packages, data analysis pipelines
January 2024	-	May 2024	Finalizing data analyses, collaborations
June 2024	-	May 2025	Finalizing collaborations, writing manuscripts and publishing

2.2 Objectives

The primary objective of the proposed interdisciplinary project is to investigate the neural, behavioral and computational mechanisms underlying the development of respiratory interoceptive models. We aim to provide a comprehensive explanation on how attention to respiratory interoceptive information modulates neural and bodily processes in children. Grounding our hypotheses on the predictive processing framework, we will examine whether and how active modulation of the breath minimizes short-term and long-term prediction errors through active inference (i.e. changing the breathing rhythm). With an interdisciplinary developmental approach combining cutting-edge neuroscientific methods with robotics

simulations, we will provide a mechanistic explanation on how sense of self, operationalized through respiratory interoceptive sensitivity, continues to develop in childhood. Below, we describe the main objectives of the three work packages addressed in the current proposal.

In the first work package, we aim to run a thorough investigation into respiratory interoception in children. We will examine how attention to respiratory interoceptive signals modulates neural and behavioral dynamics in the developing brain. This work package will lay the foundation for further work packages. In the first study, we will measure the instantaneous respiration modulated neural oscillations (RMNOs) in the developing brain. We will simultaneously record neural and respiratory signals during a resting state phase. We also plan to monitor the cardiac activity for additional information on how other interoceptive signals interact with respiratory and neural processes at rest. In the second study, we will further investigate RMNOs in an interoceptive attention task. We will develop a paradigm to measure whether children can distinguish their own respiratory signals as an indication of respiratory interoceptive attention. These studies will provide groundbreaking insights on the respiratory interoception in the developing brain.

In the second work package, we will investigate the neural, behavioral and computational mechanisms underlying active modulation of respiratory interoceptive signals. With an interdisciplinary approach, we will examine whether and how slow-paced breathing alters interoceptive predictions, and hence, interoceptive generative models. In the first study, we will use a biofeedback procedure to investigate RMNOs and behavior as children perform a respiration-based sensorimotor learning task, namely, slow-paced breathing. This procedure will allow us to examine how the breath, as an actively malleable interoceptive modality, modulates neural and behavioral processes. In the second study, together with our collaboration partner Dr. Guido Schillaci, who is an affiliated PI in the Priority Program, we will model the learning of slowing down the breath as an active inference mechanism. Based on the Predictive Processing framework, we will formalize the active modulation of the breath and the emotional experience in terms of short-term and long-term prediction error minimization. This will be the first-ever work investigating the computational mechanisms of active modulation of respiratory interoceptive signals in development.

In the third work package, our goal is to examine the prolonged effects of repeated slow-paced breathing on neurorespiratory dynamics as well as on interoceptive attention in children. We will run a 30-days training study in which children will repeatedly practice slow-paced breathing using a biofeedback procedure. The measurements in the first work package, which examine the dynamic coupling of neural and respiratory signals at rest and in an interoceptive attention task, will serve as pretest. After the training, we will run the measurements again as a posttest procedure. This research will allow us to elucidate the long-term effects of active modulation of respiratory interoceptive signals on the neural, bodily and interoceptive processes in young children. Moreover, it will provide valuable insights into whether and how interoceptive skill training alters interoceptive generative models in the developing brain. Results of this work can particularly be informative for designing complementary intervention methods for children with interoceptive and emotional challenges.

2.3 Work programme including proposed research methods

2.3.1. Work package 1

In work package 1, we will investigate the dynamic alignment of neural and respiratory signals at rest and in an interoceptive attention task. For both studies, the participants will sit in front of a monitor and will be asked to breathe as usual without holding or forcing their breaths. The participants will be instructed to inhale with their noses, as it has been shown that only nostril breathing generates phase-entrained oscillations both in humans and in mice (Ito et al., 2014; Zelano et al., 2016). The measurements in this work package will serve as pretest measures for the experiments in work package 3.

2.3.1.1. Respiration modulated neural oscillations at rest

Participants

We will test 6- to 8-year-old children. We will compare the data of younger and older children in all of our analyses in the project to account for developmental differences. According to a priori power-analyses, to detect an effect with an effect size of .3 with a power of 0.95, we will have to test a total of 40 children. Given the attrition rates in developmental neuroscience studies, we plan to test around 80 children (for 2 to 3 times). Participants will be recruited from a database of volunteer families. For more information on participants, please see sections 2.3.2.1., 2.3.3.1 and 5.1.2.

Method

Same as the set-up in the first phase of the project, in order to obtain the respiration data, we will use a respiratory belt transducer surrounding the child's upper abdomen (ADInstruments, www.adinstruments.com), which detects the thoracic changes in the circumference during respiration. At the same time, three disposable AgCl electrodes will be attached to the participant's chest in a 3-lead ECG set-up, which will allow us to monitor their cardiac activity. Both the respiratory belt and the ECG set-up will be connected to the recording hardware, which synchronizes the two signals (PowerLab, ADInstruments). Using the software provided by the manufacturer (LabChart, ADInstruments), we will continuously monitor the participant's cardiac and respiratory characteristics shown on the interface. The EEG data will be recorded via a mobile EEG system (Smarting, mBrainTrain, Belgrade, Serbia) made up of 24 Ag/AgCl electrodes, with AFz as ground and FCz as reference, placed on an EEG cap (EASYCAP, Herrsching, Germany). A small wireless, head-mounted amplifier will be attached to the back of the cap that sends digitized EEG data via Bluetooth to a receiver dongle. The Bluetooth dongle will be connected to a laptop running SmartingLSL, which records the data with a sampling rate of 500 Hz. Similar to the procedures implemented in the first phase of the Priority Program, we will use the LSL library to synchronize the neural signals with cardiac and respiratory signals.

In the first study, we will simultaneously measure children's neural and respiratory activity during a resting state phase as they watch a calming video of an undersea environment sitting in a relaxed position (in case supine rest will not be feasible). We will exclude the first 30 seconds from the analyses because breathing rates stabilize within that time (Herrero et al., 2018). The stimulus presentation, operated in MATLAB, will last 10 minutes with a pause after the first 5 minutes. The measurements will be recorded and coded offline to eliminate trials in which children perform mouth breathing. The experiments in this work package will be conducted by the main applicant's team.

Measures

In the first phase of the Priority Program, we implemented a procedure to measure how cardiac activity modulates neural processes such as heartbeat evoked potentials. In the second phase, we will focus on the role of respiration on neural processes by investigating respiration modulated neural oscillations (RMNOs). To investigate to what extent rhythmic neural activity is coupled to natural breathing at rest, we will use phase-amplitude coupling (PAC) measures. PAC measures will allow us to quantify the extent to which the amplitude envelopes of neural oscillations at certain frequencies are modulated by the phase of respiration. Considering the challenges in developmental EEG regarding data length and signal-to-noise-ratio, we will calculate phase-amplitude coupling using modulation index (MI) measure, which is shown to be robust against noisy data and short epoch durations (Hülsemann et al., 2019; Tort et al., 2010). As the respiration modulated oscillations depend on the instantaneous breathing rate and young children might take up to 25-30 breaths per minute, we will examine RMNOs at several frequencies across the entire brain during a resting state period (Kluger & Gross, 2020b).

To calculate the modulation index, we will follow the approach pioneered by Tort and colleagues (2008; 2010). We will first use Hilbert transform to extract the instantaneous phase of respiration time series and the time series of the amplitude envelope, based on which a composite time series is constructed. The phases (i.e. from -180° to 180° of angle) will then be divided into 18 bins of 20° and the mean amplitude in each phase bin will be calculated. The mean amplitude in each phase bin will be normalized so that its sum over all phase bins is equal to 1, which generates a phase-amplitude distribution. Shannon entropy, which represents the amount of information of a parameter, will be computed for each phase bin. If the mean amplitude in each phase bin is equal, corresponding to a uniform distribution, Shannon entropy is at maximum (i.e. less predictability). By using the Kullback–Leibler divergence formula, one can calculate how much a given distribution deviates from the uniform distribution. The more deviation there is from the uniform distribution, the more phase-amplitude coupling, hence, higher MI values are observed. An MI value of 0 indicates lack of phase-amplitude coupling (Hülsemann et al., 2019; Tort et al., 2008; Tort et al., 2010).

To control for spurious coupling effects, we will generate 200 surrogate modulation indexes by using a phase shuffling approach in which we randomly shift the respiratory phase time series, which will be compared against the original MI values (Kluger & Gross, 2020b; Tort et al., 2008). We will perform cluster-based permutation tests as commonly used in the literature for statistical significance testing (Maris & Oostenveld, 2007). If the clusters based on the original MIs show a larger test statistic than 95% of the randomized surrogate data, they will be considered as significant. We will run a false discovery rate (FDR) correction across frequencies at a significance level of $\alpha = .05$ to control for multiple comparisons (Kluger & Gross, 2020b).

Moreover, we will examine overall alpha power at rest, which will serve as a pretest measurement for the third work package. As the cardiac activity is known to be coupled to respiratory and neural processes, we will also monitor the cardiac activity. We will calculate commonly investigated measures such as HRV and RSA to use as complementary measures to assess the efficacy of slow-paced breath training in work package 3.

Hypotheses

Previous research in adults has shown that the amplitude of global oscillatory brain activity is significantly modulated by respiration in a broad frequency range from 2 to 150 Hz at rest (Kluger & Gross, 2020b). We expect to find a similar respiration-locked modulation indicated by a high normalized modulation index across a broad frequency spectrum over the entire brain in children. As the nature of respiration modulated neural oscillations in developmental populations is entirely unknown, predictions have to be tentative. Our work will be highly informative to determine whether natural breathing modulates oscillatory brain activity in children similar to the emerging evidence in adults (e.g. Hsu et al., 2020; Kluger & Gross, 2020b; Zelano et al., 2016).

2.3.1.2. Respiration modulated neural oscillations in the interoceptive attention task

Participants

The same participants (N=80) as in the first experiment will be tested with the interoceptive attention paradigm. Following the resting state measurements, the participants will continue performing the experiment on respiratory interoceptive attention. We will use a paradigm similar to our previous work with the IBREATH paradigm as well as the work by Wang and colleagues (2019).

Method

In this task, we will again use an undersea environment as the background. On top of this background, we will present a visual display (e.g. adjacent starfish) positioned on a continuous sine wave, in other words, a breath curve, oscillating from left to the right of the screen. As described in section 2.3.1.1., the EEG data will be recorded simultaneously to the

respiratory data. The ECG data will be obtained concurrently and used to control for the effects of cardiac fluctuations (Hsu et al., 2020).

In the interoceptive attention task, we will measure children's attention to their respiratory signals. The task will consist of two types of trials. *In synchronous trials*, participants will observe a breath curve that maps onto their actual respiratory rhythm. *In the asynchronous trials*, the stimulus presentation algorithm will generate a respiration-like rhythm that is faster or slower than the same child's average respiration rate recorded from the trial before. The task of the participant will be to indicate whether the breath curve is in sync or out of sync with their own breathing tempo. We will semi-randomize the presentation order of the fast and slow versions of the asynchronous trials. The order of the synchronous and asynchronous trials will also be semi-randomized to make sure that participants do not try to entrain their breathing rhythms to asynchronous presentations, although previous research with adults did not report such effects (Wang et al., 2019). In some trials, a puffer fish image will flash at a random position on the breath curve (i.e. on the starfish). We will explain the purpose of this stimulation in the next part. We will decide on the final form of the paradigm after the pilot measurements. In case the puffer fish would be too distracting for the children, we will only use this stimulation in some of the trials in the interoceptive attention task and discard those trials in the analyses.

To disentangle interoceptive attention from exteroceptive attention or general arousal, participants will perform an exteroceptive attention task as a control condition. **In the exteroceptive attention task**, the participants will again observe a display representing a breath curve that is either in sync or out of sync with their respiration. Similarly, in some trials, a puffer fish image will flash at a random position on the breath curve. Here, the task of the participants will be to indicate whether a puffer fish appeared on the breath curve or not. The task will be adjusted in difficulty so that there will be incorrect answers.

Each trial will last 20 seconds with an average inter-trial-interval of 2 seconds. The trial will start with an undersea world display that lasts for 5 seconds. This will be followed by a 10 second presentation of the breath curve oscillating either in sync or out of sync with the participant's breathing rhythm. At the end of the trial, a 5 second response window will be presented during which the participants will use a button press to indicate whether the breath curve was in sync or out of sync with their respiration rhythm or whether a puffer fish flashed on the screen or not, depending on the task demands. The experiment will consist of 60 trials in total and last 12 minutes. The experiments in this work package will be conducted by the main applicant's team.

Measures

To analyze the respiration modulated neural oscillations in the interoceptive attention task, we will use the modulation index measure as described in detail in section 2.3.1.1. We will calculate the MIs for correct and incorrect responses for each trial type in the interoceptive attention task. We will run the same analyses for the control condition measuring exteroceptive attention. We will obtain an MI difference score (MI_{DS}) as the absolute proportion of the difference between modulation index in the correct and incorrect trials. We will then test whether MI_{DS} predicts behavioral performance.

For behavioral analyses, we will quantify performance accuracy, discrimination sensitivity and response bias based on the approaches used in previous studies (Harrison et al., 2020; Wang et al., 2019). For both interoceptive and exteroceptive attention tasks, performance accuracy will be calculated based on the total correct responses in both trial types. Moreover, we will use signal detection theory to disentangle discrimination sensitivity (d') and response bias (β) of the participants. Response bias will be calculated as a control variable. Here, d' quantifies how sensitive one is to bottom-up input, for example, the breath, whereas β measures how willing a person is to indicate a positive response influenced by top-down processes such as expectations. We will calculate the sensitivity index as follows: $d' = Z_{hit\ rate} - Z_{false\ alarm\ rate}$. For example, in the interoceptive attention task, hit rate refers to the proportion of

trials in which the participants pressed “yes” in response to a synchronous breath curve whereas false alarm rate is the proportion of trials in which participants pressed “yes” in response to an asynchronous breath curve. A higher value of d' represents better interoceptive sensitivity, whereas a value of 0 indicates chance-level performance. As for the β , we will use the following formula: $\beta = \exp(d' \times C)$, where $C = -(Z_{\text{hit rate}} + Z_{\text{false alarm rate}})/2$. Values less than 1 indicate a bias toward responding “yes” in a trial, whereas values greater than 1 signify a bias toward the “no” response. A value of 1 indicates no bias (Stanislaw & Todorov, 1999). For the exteroceptive attention task, the same procedure will be applied in which the presence and absence of the puffer fish will indicate “signal” or “noise” (Wang et al., 2019).

Hypotheses

Previous research with adults has shown increased respiration modulated neural activity in interoceptive attention tasks (Herrero et al., 2018). Accordingly, we hypothesize that children will show higher respiration modulated oscillatory activity, as quantified by the modulation index, in correct trials in the interoceptive attention task as compared to incorrect trials and to the exteroceptive attention task. Moreover, we expect that the MI_{DS} will predict performance accuracy and discrimination sensitivity in the interoceptive attention task.

2.3.2. Work package 2

2.3.2.1. Respiration modulated neural oscillations during slow-paced breathing

Participants

We plan to perform the experiments described in work packages 1 and 2 on the same day. After the participants (N=80) complete the experiments in the first work package, they will perform the experiments described in the current work package in this specific order to avoid carry over effects. Participants will take as many breaks as they need between measurements. In case measurements of the second work package cannot be completed on the same day, participants will be given another appointment within the same week.

Method

The measurement set-up will be the same as the set-up described in detail in section 2.3.1.1. Participants will be wearing the respiratory belt, EEG cap and the ECG electrodes. The synchronization between respiratory and neural signals will be achieved via LSL. We will also obtain ECG data concurrently to record cardiorespiratory dynamics during the slow-paced breathing (please see the subproject below) and to monitor the HRV after the slow-paced breathing (Hsu et al., 2020; for a review Zaccaro et al., 2018).

We will implement a modified version of a respiratory biofeedback game used in the previous research with children (Sonne & Jensen, 2016). In this biofeedback game, the participants will control a virtual avatar (i.e. a puffer fish) in an undersea world using their breathing rhythm. The child's inhalation and exhalation activity will be measured through a respiratory belt. We will use the respiratory belt as the game controller. The logic of implementation will be similar to the iBREATH paradigm and the paradigms described in the first work package. The experiment will start and end with 2 minute **resting state conditions** in which the participants will passively observe an underwater world while their neural, cardiac and respiratory activities are recorded.

In the slow-paced breathing condition, participants will observe the starfish positioned on the screen representing a slow-paced breath wave. The goal of the participant will be to collect the starfish. The starfish will be positioned on the screen such that a slow-paced breathing pattern should be applied to collect as many starfish as possible. When the child inhales, the virtual avatar (i.e. a puffer fish) will move towards the top of the screen and at the same time expand in size. During exhalation, the puffer fish will move towards the bottom of the screen and shrink in size. If the child holds their breath, the puffer fish will not move or change

size. The condition will last 10 minutes, divided into two parts, corresponding to 60 full slow-paced breath cycles (i.e. 6 cycles per minute).

In the natural breathing condition, participants will observe starfish positioned on the screen representing their average natural breathing rate. This parameter will be defined individually for each participant. The virtual avatar will move upwards and downwards on the screen to collect the starfish based on the actual breathing rate of the participant to keep the task characteristics similar across conditions. This condition will last for 5 minutes and serve as a control condition.

The order of the natural and slow-paced breathing conditions will be counterbalanced across participants. We will not include a fast-paced breathing condition in our experiment with children in order to avoid potential negative effects associated with fast breathing such as anxiety and panic attacks (Nardi et al., 2009). We will measure the EEG, ECG and the respiratory signals continuously and segment the data offline. One trial will correspond to the upwards and downwards movement of the avatar. Each upwards and downwards movement will be marked separately and triggers will be sent to the recording software. The experiments in this work package will be conducted by the main applicant's team.

Measures

In order to investigate the effects of active modulation of the breathing tempo on neural processes, we will use modulation index measures (for more information, please see section 2.3.1.1.). Whereas it has been mostly shown that broadband gamma activity couples with respiratory cycle in mice and in humans during normal to fast breathing (Herrero et al., 2018; Ito et al., 2014; Yanovsky et al., 2016; Zelano et al., 2016), others have reported modulation of cortical phase activity in the alpha range across widespread brain areas during slow-paced breathing (Hsu et al., 2020). Based on previous findings, we will particularly focus on alpha and gamma amplitudes at different respiration phases across all regions and calculate the modulation index for each condition. Given the potential differences in the EEG activity between children and adults (Barriga-Paulino, et al., 2011), we will further extract the amplitudes in all of the other major frequency bands (e.g. delta, theta and beta) and calculate the modulation index accordingly.

Moreover, as previous studies with adults showed increased alpha band activity during and after slow-paced breathing (Fumoto et al., 2004; Park & Park, 2012; Yu et al., 2011), we will additionally examine power in the alpha frequency band during and immediately following the slow-paced breathing. It has been argued that sensory deafferentation that occurs during the slow-paced breathing brings about an attentional shift directed towards internal signals leading to an increase in the alpha activity (for a review Zaccaro et al., 2018). As a behavioral measure of learning to slow down the breath, we will use the number of points (i.e. starfish) earned in the game. The more points the child gets, the better they will perform the task, that is, they will actively slow down their breathing rhythms based on feedback.

Hypotheses

We hypothesize that neural oscillations will vary not only as a function of the phase of respiration but also as a function of the rate of respiration (i.e. natural vs. slow). That is, we expect to observe differential modulation index for volitional modulation of the breath (Herrero et al., 2018; Hsu et al., 2020). More specifically, based on previous findings in adults, we predict a reduction in MI as the breath gets slower (Hsu et al., 2020). Moreover, we expect to observe an increase in alpha power during and immediately after the slow-paced breathing. We predict a negative correlation between the overall learning outcome (i.e. number of points earned) and the MIs, as the participants learn to slow down their breaths. Given that there is no previous research on how active modulations of the breath impact neural oscillations in the developing brain, we believe the proposed research will be highly valuable to shed light on this topic.

Trial by trial adaptation is an essential component of learning. As a subproject (e.g. PhD thesis), we plan to investigate trial by trial alterations in cardiorespiratory (subproject 1), neurorespiratory (subproject 2) and behavioral responses (subproject 3) over the course of the experiment to examine the mechanisms of how children adapt to self-generated errors during learning. First, we will measure trial by trial changes in the synchronization between cardiac and respiratory functions (i.e. cardiorespiratory coupling) over the course of the experiment as participants learn to slow down their breath. Second, we will measure trial by trial adjustments in the respiration modulated neural oscillations to investigate the adaptation of neural processing during learning. Third, we will track trial by trial behavioral progress in the task by measuring the deviations between two paths (i.e. fixed slow-paced breath curve vs. participant generated respiration trajectory) using algorithms such as ALCAMP, which quantify the mapping between two trajectories (Mueller et al., 2016). This work will provide important insights on how children learn slow-paced breathing and adapt to uncertainty.

2.3.2.2. Computational mechanisms of active modulation of the breath and emotions

Breathing happens at the present moment and grounds the organism in the current sensory experience, which is the essence of a minimal self. In our work, we have argued that through active inference, that is, changing the breathing rhythm, respiration can alter interoceptive predictions, and the felt experience thereof (Boyadzhieva & Kayhan, 2020). As respiration-related sensory evidence is recontextualized, the predictions are altered: in the case of slowing down the breathing rhythm, they become aligned with a relaxed state, rather than other, possibly conflicting, interoceptive states (i.e. an anxious state). This formulation, therefore, accounts for the efficacy of a slow breathing rhythm to achieve a positive emotional and physiological state. At the same time, it acknowledges its bidirectionality, explaining how accelerated breathing rhythms can upregulate the system in a way that results in an aroused state. Nevertheless, the question of how controlled breathing acts as an active inference mechanism to alter interoceptive predictions, hence emotional experience, should be further clarified and formalized (Boyadzhieva & Kayhan, 2020).

Together with our cooperation partner, Dr. Guido Schillaci, we plan to conduct robotics experiments to investigate the computational mechanisms of active modulation of the breath and associated emotional experience. We will formalize the learning to slow down the breath as a prediction error minimization process, as defined in the Predictive Processing framework (Friston, 2010). We hypothesize that slowing down the breath will be correlated to a descending trend in the PE dynamics, which are associated with positive physiological and emotional outcomes (i.e. a calm state and pleasure). On the contrary, ascending PE trends will be associated with increased movement and negative emotional valence (i.e. rapid breathing and distress). We believe that our interdisciplinary work will not only be of great interest to researchers investigating the developing interoceptive brain but it will also benefit developmental roboticists aiming at formalizing the exact computational mechanisms of interoceptive and emotional processes.

We will first synchronize our work with existing robotic studies conducted by Schillaci and colleagues (2020) modeling how regulating behavior interacts with prediction error dynamics and learning performance. We will then design a robotics experiment in which the robot performs a specific learning task (i.e. earning points by slowing down movements). We will formalize a regulatory mechanism in which the rhythm and velocity of robot's movements (e.g. slow vs. fast) influence short-term PE dynamics, hence overall learning and emotional outcomes (Schillaci et al., 2020). Moreover, we plan to extend our work to long-term updating of predictive capabilities testing the influence of memory consolidation, as we aim to understand whether repeated experience with active modulation of the breath can also elicit long-lasting changes in interoceptive generative models (Boyadzhieva & Kayhan, 2020). Accordingly, findings obtained from experiments in work package 3 of the proposed project will inform the formalization of long-term model updating dynamics. More information on the computational details of the studies can be found in Dr. Guido Schillaci's proposal titled "Predictive Processes and Self-Regulatory Mechanisms for a Developing Robot".

2.3.3. Work package 3

2.3.3.1. Neurophysiological effects of respiratory interoception training

Participants

During the initial call for a testing appointment, all participants will be informed about the sessions that will be conducted outside of the laboratory (i.e. in their homes). The families will be given the option to install the training games/apps to their smartphones on their own or with the help of our team in the laboratory. At the end of their first visit, the participants (N=80) will be randomly assigned to either the training group or the active control group. We will invite the participants for another visit to complete posttest measurements.

Method

For the training group, we will use mobile based slow-paced respiration training games such as Breath Pacer or Calm, which enable players to perform short sessions of slow breathing. For example, the Calm game consists of 12 “breather clouds” on the mobile screen, which instructs the player to take a deep breath in (from the nose) and exhale into the microphone of the mobile phone to dissipate clouds. As the clouds disappear by consecutive slow breaths, a calming scene appears on the screen. The game is by default calibrated to a cycle of 6 breaths per minute (Agrawal et al., 2020). **For the active control group**, we will use free apps such as Ocarina 2 or equivalent whereby players blow into the microphone of their smartphones and press on the buttons that appear on the screen to produce flute-like sounds. The player earns “breath points” as they play more songs.

Both groups will practice with the games for 15 minutes every day for 30-days under the supervision of their caregivers. We are aware that although a 30 days period is a common practice in training research (Laborde et al., 2019), we might have to reduce training duration in order to avoid drop-outs and increase the feasibility of the procedure for the families. The final decision on the duration of the training will be made based on the pilot testing. The families will be asked to record the sessions and fill in a training diary, which will be uploaded to a secure platform to be shared with the experimenters on a daily basis. If no entry was made in the online platform, the families will receive reminders to continue with the procedure. Both groups will then be invited to the lab for the posttest measurements in the week after the completion of the training.

Measures

We will repeat the same measurements that are described in work package 1 for posttest measurements. For detailed information, please see sections 2.3.1.1. and 2.3.1.2. We will also use parental questionnaires to evaluate overall development of the children, particularly focusing on motor development, as studies show that gross motor functions correlate with pulmonary and respiratory muscle performance in atypically developing children (Kwon & Lee, 2014). We will also use child-friendly self-report questionnaires (e.g. Child Anxiety Meter) to monitor anxiety and other psychological factors.

Hypotheses

Research has shown that breath training modulates physiological and neural processes in healthy adults (Laborde et al., 2019; for a review, Lomas et al., 2015). As there has been no research with children measuring neural dynamics following a slow-paced breath training procedure, we believe findings obtained from this work package will be highly informative. It could be that because instantaneous modulation index decreases when the breath deepens and slows down (Hsu et al., 2020; Kluger & Gross, 2020a) and experienced breath meditators naturally breathe slower and deeper as compared to novices (Farb et al., 2013), we will observe an overall reduction in the MI scores at rest in the posttest phase, if the training induces prolonged changes in the breathing patterns. As previous research has shown high overall

alpha activity during and after slow-paced breathing (for a review Zaccaro et al., 2018) as well as in mindfulness practitioners (for a review Lomas et al., 2015), we expect to observe increased power in the alpha band at rest following the training. We will also explore whether slow-paced breath training leads to a general increase in HRV and RSA values at rest, as shown in adults (for a review, Zaccaro et al., 2018). Moreover, we expect the participants to perform better in the interoceptive attention task in the posttest phase as compared to the pretest, if the training enables them to more readily detect their respiratory interoceptive signals (Farb et al., 2013). Accordingly, we expect to observe increased neural activity, higher performance accuracy and discrimination sensitivity scores in the respiratory interoceptive attention task following the breath training.

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4 Relevance of sex, gender and/or diversity

We will record the information about the sex of the participant. We will try to include equal numbers of boys and girls in the sample.

5 Supplementary information on the research context

5.1 Ethical and/or legal aspects of the project

5.1.1 General ethical aspects

We will conduct psychophysiological and neural recordings with children at 6 to 8 years of age. These methods are non-invasive and widely used by developmental researchers across the world.

5.1.2 Descriptions of proposed investigations involving experiments on humans or human materials

The participants will be recruited on a voluntary basis from a database of participants at the University of Potsdam. Participant data will be saved in an anonymous format and written consent forms will be obtained from the parents prior to the experiments in accordance with the GDPR regulations. All involved parties will be debriefed about the aim of the experiments. We will obtain ethical approval from the University of Potsdam Ethics Board before conducting the experiments.

5.1.3 Descriptions of proposed investigations involving experiments on animals

N/A

5.1.4 Descriptions of projects involving genetic resources (or associated traditional knowledge) from a foreign country

N/A

5.1.5 Descriptions of investigations involving dual use research of concern, foreign trade regulations

N/A

5.2 Data handling

Electronic storage of data will be ensured in accordance with DGPs/APA guidelines. All personal data will be assessed, processed, and stored in accordance with the EU General Data Protection Regulations. Personal data (address, name, etc.) will be treated as strictly confidential. Only the researchers involved in the study and their scientific collaborators will have access to the research data and will be obliged to treat the data confidentially. Recordings of the child will be stored without a name. The participants will be given a unique participant ID (e.g. k4092). The unique participant ID will be stored in the internal database of the Developmental Psychology Unit at the University of Potsdam, which will be strictly protected. On a password protected file, this participant ID number will be listed together with the participant number that is used to save the participant data (e.g. P01). Only the researchers of the study will have access to this password protected file. The digital data will be saved for 10 years after the final publication and then deleted. Within these 10 years, the participants can request the deletion of their data from the study management using their individual participant ID number at any time and without giving reasons. All paper records from the experiments will be stored in a locked filing cabinet, and will be destroyed after 5 years. Upon request, only the fully

anonymized data and data-analysis flow-charts will be made accessible for other researchers and for publications.

5.3 Other information

We planned further collaborations with other researchers in the Priority Program, namely, Prof. Dr. Markus Raab and Dr. Lisa Musculus, on a theoretical paper on interoception from a developmental embodied cognition perspective. The main applicant, Dr. Ezgi Kayhan Wagner, is the co-editor of the Frontiers in Psychology Research Topic “The Mechanisms Underlying the Human Minimal Self” together with other members of the Priority Program, namely Prof. Dr. Bernard Hommel, Prof. Dr. Verena Hafner, Prof. Dr. Markus Paulus, Prof. Dr. Dongheui Lee and Dr. Stephan Verschoor. The main applicant Dr. Ezgi Kayhan Wagner has served as the Equal Opportunities Representative in the first phase of the Priority Program (succeeded by Dr. Laura Kaltwasser).

6 People/collaborations/funding

6.1 Employment status information

The main applicant Dr. Ezgi Kayhan Wagner is a postdoctoral researcher funded through a “Temporary Positions for Principal Investigators” grant awarded to her within the first phase of the DFG Priority Program “The Active Self”. The main applicant has a three year fixed-term contract in the University of Potsdam that ends on 30.04.2022.

6.2 First-time proposal data

N/A

6.3 Composition of the project group

Dr. Guido Schillaci, postdoctoral researcher, full-time, fixed
Prof. Dr. Birgit Elsner, Professor, full-time, permanent
Prof. Dr. Stefanie Hoehl, Professor, full-time, permanent

6.4 Researchers in Germany with whom you have agreed to cooperate on this project

The research will be conducted in the University of Potsdam using the lab facilities provided by Prof. Dr. Birgit Elsner. The declaration letter of the host is attached to the application.

6.5 Researchers abroad with whom you have agreed to cooperate on this project

For the robotics simulations, the main applicant Dr. Ezgi Kayhan Wagner agreed on a collaboration with developmental roboticist Dr. Guido Schillaci of The BioRobotics Institute, Department of Excellence in Robotics & AI, Scuola Superiore Sant’Anna. Prof. Dr. Stefanie Hoehl will provide consultancy on the experiments with children and on data analyses upon request. The cooperation agreement letters are attached to the application.

6.6 Researchers with whom you have collaborated scientifically within the past three years

Prof. Dr. Manos Tsakiris, Prof. Dr. Harold Bekkering, Prof. Dr. Sabine Hunnius, Prof. Dr. Markus Paulus, Dr. Jill O’Reilly, Dr. Robert Oostenveld, Dr. Sam Wass, Prof. Dr. Verena Hafner, Prof. Dr. Stefan Wermter, Dr. Phuong Nguyen, Dr. Manfred Eppe, Prof. Dr. Yukie Nagai, Dr. Iris van Rooij, Dr. Johan Kwisthout, Dr. Lieke Heil, Dr. Marlene Meyer, Dr. Moritz Köster, Dr. Emma Ward, Dr. Radoslaw Cichy, Prof. Dr. Natalie Boll-Avetisyan, Dr. Christine

Michel, Dr. Miriam Langeloh, Dr. Pascal Vrtička, Dr. Hanna Schleihauf, Dr. Melanie Kungl, Prof. Dr. Sabien Pauen.

6.7 Project-relevant cooperation with commercial enterprises

N/A

6.8 Project-relevant participation in commercial enterprises

N/A

6.9 Scientific equipment

We will continue using the PowerLab with LabChart software, the ECG set-up and the respiratory belt for the psychophysiological measurements. Smarting mobile EEG systems will be used for the EEG recordings. The equipment was purchased in the first phase of the Priority Program.

6.10 Other submissions

N/A

7 Requested modules/funds

7.1 Basic Module

7.1.1 Funding for Staff

Given the intensity of the data collection procedures in developmental populations, the abundance of the methods and the complexity of the data analyses approach, the project will require a student assistant (i.e. 40 hours per month) and a PhD student (Psychology, 75%, 3 years) to realize the experiments together with the main applicant Dr. Ezgi Kayhan Wagner. Based on previous experience, we first plan to hire a student assistant with a computer science background for 8 months during the implementation and piloting period. We then plan to hire another student assistant with a psychology background for the rest of the funding period (i.e. 28 months) to help the PhD student and the main applicant Dr. Ezgi Kayhan Wagner with the data collection, data coding and other organizational aspects of the project. We plan to get additional support from master's students in the University of Potsdam. Accordingly, the project requires 24,463 Euros for the student assistant salary and 157, 275 Euros for the PhD student salary. A total cost for funding the staff is **181, 738 Euros**.

7.1.2 Direct Project Costs

7.1.2.1 Equipment up to € 10,000, Software and Consumables

We will use the same equipment purchased in the first phase of the Priority Program, which reduces our equipment costs greatly. However, because we will test an older age group in the current project, we will need one more EEG cap (i.e. EASYCAP 24Ch Real-Behaviour EEG-Recording Cap for Smarting with 24 built in electrodes according to the 10-20 system) in size 52 (800 Euros). Together with EEG application gels, disposable ECG electrodes and other consumables, we will need **2000 Euros** for the equipment costs.

7.1.2.2 Travel Expenses

Results from this project will be presented at international conferences at least once per year. In case the conferences take place physically, we will need a total of **3000 Euros** for conference travel expenses.

7.1.2.3 Visiting Researchers (excluding Mercator Fellows)

N/A

7.1.2.4 Expenses for Laboratory Animals

N/A

7.1.2.5 Other Costs

We will give children books or toys to the participants in return of their participation. An amount of **3000 Euros** will be needed for participant payments in accordance with reimbursement rates of the University of Potsdam BabyLab.

7.1.2.6 Project-related Publication Expenses

We believe that the proposed interdisciplinary project will generate groundbreaking results, which will lead to a series of publications. In accordance with the DFG regulations, we request a total of **2,250 Euros** for open access publication fees. We will request additional funding from the University of Potsdam to cover the rest of the publication costs.

7.1.3 Instrumentation

7.1.3.1 Equipment exceeding € 10,000

N/A

7.1.3.2 Major Instrumentation exceeding € 50,000

N/A

7.2 Module Temporary Position for Principal Investigator

The main applicant, Dr. Ezgi Kayhan Wagner, will fund her research full-time based on the module for "Temporary Positions for Principal Investigators" for 36-months. The salary level will be based on the DFG personnel rates (i.e. post-doctoral researcher or comparable: E14 Level 2 100 %). The main applicant Dr. Ezgi Kayhan Wagner will realize the project, more specifically, lead the team, set-up the experiments, implement the paradigms, train and supervise the staff, oversee the measurements, prepare data analyses scripts, analyze the data, establish collaborations, present the work to the scientific community, participate in outreach activities and publish papers. An amount of **226, 800 Euros** will be needed for funding this position.

7.3 Module Replacements

N/A

7.4 Module Temporary Substitute for Clinicians

N/A

7.5 Module Mercator Fellows

N/A

7.6 Module Project-Specific Workshops

N/A

7.7 Module Public Relations

N/A

7.8 Module Standard Allowance for Gender Equality Measures

N/A